

The Schur Expansion of the Coinvariant Frobenius at a Root of Unity:

Support, Signs, and the Virtual-Character Obstruction

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Abstract

Let $n \geq 1$, $e \geq 2$, and write $n = ek' + r_0$ with $0 \leq r_0 < e$. Let $q_e^{(n)} := \sum_{\lambda \vdash n} \tilde{f}_\lambda(\zeta_e) s_\lambda$, where $\tilde{f}_\lambda(t)$ is the fake degree of λ (the graded multiplicity of V_λ in the coinvariant algebra of S_n) and ζ_e is a primitive e -th root of unity. Combining the factorisation $q_e^{(n)} = p_e^{k'} \cdot q_e^{(r_0)}$ (Clio 2026, [1]) with the skew form of Farahat's theorem on characters at e -regular classes, we prove that the Schur coefficients of $q_e^{(n)}$ have the closed form

$$[s_\lambda]q_e^{(n)} = d_{\mu_0}(\zeta_e) \cdot \varepsilon_e(\lambda) \cdot \binom{k'}{k_0, \dots, k_{e-1}} \cdot \prod_{i=0}^{e-1} f^{\lambda_{(i)}},$$

where μ_0 is the e -core of λ (necessarily of size r_0), $\lambda_{(0)}, \dots, \lambda_{(e-1)}$ are the e -quotients with sizes $k_i := |\lambda_{(i)}|$, f^ν is the dimension of the Specht module V_ν , $\varepsilon_e(\lambda) \in \{\pm 1\}$ is Farahat's border-strip parity sign (well-defined by James–Kerber §2.7), and $d_{\mu_0}(\zeta_e) = [s_{\mu_0}]q_e^{(r_0)} \in \mathbb{Z}[\zeta_e]$. As a consequence $q_e^{(n)}$ has mixed Schur signs for every (n, e) with $k' \geq 1$ (a genuine regular-representation obstruction), so no submodule of any $\mathbb{Z}[\zeta_e]$ -graded S_n -module realises $q_e^{(n)}$ as its graded Frobenius image. This upgrades §6 of the composite- d paper from a queued frontier to a proved structural theorem.

1 Setup and Prior Factorisation

Fix $n \geq 1$ and let Λ_n be the space of homogeneous symmetric functions of degree n over $\mathbb{Q}$. For a partition $\lambda \vdash n$, the *fake degree* of λ is

$$\tilde{f}_\lambda(t) = \sum_{T \in \text{SYT}(\lambda)} t^{\text{maj}(T)} = t^{n(\lambda)} \frac{\prod_{i=1}^n (1 - t^i)}{\prod_{c \in \lambda} (1 - t^{h(c)})},$$

which equals the graded multiplicity of the Specht module V_λ in the coinvariant algebra H_n of S_n . Fix $e \geq 2$ and let $\zeta_e := e^{2\pi i/e}$. Set

$$q_e^{(n)} := \sum_{\lambda \vdash n} \tilde{f}_\lambda(\zeta_e) s_\lambda \in \Lambda_n \otimes \mathbb{Z}[\zeta_e]. \tag{1}$$

Throughout write $n = ek' + r_0$ with $0 \leq r_0 < e$, and denote by $q_e^{(r_0)}$ the analogous specialisation at rank r_0 : $q_e^{(r_0)} = \sum_{\mu \vdash r_0} \tilde{f}_\mu(\zeta_e) s_\mu \in \Lambda_{r_0} \otimes \mathbb{Z}[\zeta_e]$. Since $r_0 < e$, every hook length of every $\mu \vdash r_0$ is at most $r_0 < e$, so $\tilde{f}_\mu(\zeta_e) \neq 0$ for every $\mu \vdash r_0$. In particular $q_e^{(r_0)} \neq 0$ whenever $r_0 \geq 1$.

Proposition 1 (Coinvariant Verschiebung, [1]). *For all $n \geq 1$ and $e \geq 2$,*

$$q_e^{(n)} = p_e^{k'} \cdot q_e^{(r_0)} \quad \text{in } \Lambda_n \otimes \mathbb{Z}[\zeta_e].$$

Sketch. Both sides expand in the power-sum basis as $\sum_{\rho} c_{\rho} p_{\rho}$ with $c_{\rho} = z_{\rho}^{-1} \Psi_{\rho}(\zeta_e)$, where $\Psi_{\rho}(t) = \prod_{i=1}^n (1 - t^i) / \prod_{j \in \rho} (1 - t^j)$ is the Chevalley–Molien polynomial. The vanishing $\Psi_{\rho}(\zeta_e) = 0$ unless ρ contains exactly k' parts equal to e collapses the sum to $\rho = (e^{k'}, \nu)$ with $\nu \vdash r_0$, giving the product structure. Full proof in [1]. $\square$

Our target is the Schur expansion

$$q_e^{(n)} = \sum_{\lambda \vdash n} c_{\lambda}^{(e,n)} s_{\lambda}, \quad c_{\lambda}^{(e,n)} \in \mathbb{Z}[\zeta_e]. \quad (2)$$

2 e -Cores, e -Quotients, and Farahat's Sign

We recall the abacus bijection between partitions and pairs (core, quotient); our reference is James–Kerber [4, §2.7].

Beta numbers. For $\lambda \vdash n$, pad to length divisible by e (say length N) by appending zero rows, and set $\beta_i(\lambda) = \lambda_i + (N - i)$ for $i = 1, \dots, N$. The set $B_{\lambda} = \{\beta_1, \dots, \beta_N\} \subset \mathbb{N}$ has N elements and determines λ up to the padding choice.

e -abacus. Distribute the beads at positions B_{λ} into e runners according to residue mod e . On runner $r \in \{0, 1, \dots, e-1\}$ sit the beads at positions $\{\beta \in B_{\lambda} : \beta \equiv r \pmod{e}\}$; let k_r be their count, and their positions divided by e (equivalently: $(\beta - r)/e$) form a strictly decreasing sequence $q_1^r > q_2^r > \dots > q_{k_r}^r \geq 0$ giving the beta numbers of a partition $\lambda_{(r)} := (q_1^r - (k_r - 1), q_2^r - (k_r - 2), \dots, q_{k_r}^r - 0)$ after stripping zero rows. The tuple $\text{quot}_e(\lambda) := (\lambda_{(0)}, \lambda_{(1)}, \dots, \lambda_{(e-1)})$ is the e -quotient.

e -core. Slide every bead on every runner as far up its runner (i.e., towards smaller positions) as possible without collision, then read off the partition determined by the resulting beta-set. This yields the e -core $\text{core}_e(\lambda)$, characterised as the unique partition obtainable from λ by iterated removal of border strips of size e ; equivalently, a partition with no hook of length divisible by e .

Size relation. $|\lambda| = |\text{core}_e(\lambda)| + e \sum_{i=0}^{e-1} |\lambda_{(i)}|$. In our setup $|\lambda| = n$ so

$$\sum_{i=0}^{e-1} |\lambda_{(i)}| = \frac{n - |\text{core}_e(\lambda)|}{e}, \quad (3)$$

which is an integer iff $|\text{core}_e(\lambda)| \equiv n \pmod{e}$, i.e., $|\text{core}_e(\lambda)| = r_0$ (since $|\text{core}_e(\lambda)| < e$ and $r_0 < e$).

Farahat's sign. Farahat [3] proved that if λ has empty e -core, then the value of the character χ^{λ} at the class of e -cycles $(e, e, \dots, e) \in S_{ek'}$ is

$$\chi_{(e^{k'})}^{\lambda} = \varepsilon_e(\lambda) \cdot \binom{k'}{k_0, k_1, \dots, k_{e-1}} \cdot \prod_{i=0}^{e-1} f^{\lambda_{(i)}}, \quad (4)$$

with $k_i := |\lambda_{(i)}|$ and $\varepsilon_e(\lambda) \in \{\pm 1\}$. Equivalently, $\varepsilon_e(\lambda) = (-1)^{\sum \text{ht}(B)}$ where the sum is over the k' border strips B of any border-strip tableau of shape λ with all strips of size e ; James–Kerber [4, Thm. 2.7.32] show that this parity is independent of the tableau. Concretely, $\varepsilon_e(\lambda)$ is the sign of the permutation on B_λ that implements the bead-sliding to the e -core.

For $\mu_0 \subseteq \lambda$ with μ_0 an e -core, we extend ε_e to a sign $\varepsilon_e(\lambda/\mu_0) \in \{\pm 1\}$ by the same recipe: for any border-strip tableau of shape λ/μ_0 with all strips of size e , set $\varepsilon_e(\lambda/\mu_0) = (-1)^{\sum \text{ht}(B)}$; independence of tableau is again [4, §2.7]. When μ_0 is itself an e -core we in fact have $\varepsilon_e(\lambda/\mu_0) = \varepsilon_e(\lambda)$ (both compute the parity of the same bead-sliding permutation, since sliding beads that are already in place contributes trivially); we use the shorter notation $\varepsilon_e(\lambda)$ throughout.

3 Skew Farahat with a Multinomial Weight

The engine of the proof is the skew analogue of (4) applied to a shape λ/μ_0 whose e -quotient (of λ) accounts for all the ek' removed cells.

Proposition 2 (Skew Farahat with multinomial). *Let μ_0 be an e -core and $\lambda \supseteq \mu_0$ with $|\lambda| - |\mu_0| = ek'$. Then the skew Murnaghan–Nakayama character*

$$\chi_{(ek')}^{\lambda/\mu_0} := [s_\lambda](p_e^{k'} \cdot s_{\mu_0})$$

vanishes unless $\text{core}_e(\lambda) = \mu_0$; and if it does not vanish, then

$$\chi_{(ek')}^{\lambda/\mu_0} = \varepsilon_e(\lambda) \cdot \binom{k'}{k_0, k_1, \dots, k_{e-1}} \cdot \prod_{i=0}^{e-1} f^{\lambda_{(i)}}. \quad (5)$$

Proof. The skew Murnaghan–Nakayama rule (Stanley [6, Cor. 7.17.5]) gives

$$\chi_{(ek')}^{\lambda/\mu_0} = \sum_T (-1)^{\sum_j \text{ht}(B_j)},$$

where T ranges over sequences of shapes $\mu_0 = \lambda^{(0)} \subset \lambda^{(1)} \subset \dots \subset \lambda^{(k')} = \lambda$ with each skew shape $B_j := \lambda^{(j)}/\lambda^{(j-1)}$ a connected border strip of size e .

Vanishing. Adding a border strip of size e moves exactly one bead on the e -abacus one step down its runner (from position p to position $p+e$), which preserves the multi-set of beads on each runner. So λ and μ_0 have the same e -core, i.e., $\text{core}_e(\lambda) = \text{core}_e(\mu_0) = \mu_0$. Conversely if $\text{core}_e(\lambda) \neq \mu_0$ no such tableau exists, so the sum is empty and (5) vanishes.

Nonvanishing formula. Assume $\text{core}_e(\lambda) = \mu_0$. Adding an e -border-strip to a shape ν that slides the unique bead on runner r one step down its runner is equivalent, in the quotient partition $\nu_{(r)}$, to adding a single cell. The corresponding *quotient border-strip tableau* is thus an e -tuple $(S_0, \dots, S_{e-1})$ where each S_i is a standard Young tableau of shape $\lambda_{(i)}$ (with $|\lambda_{(i)}| = k_i$), together with a total order on the $k' = \sum k_i$ cells specifying the linear order in which they were added by the strips $B_1, \dots, B_{k'}$.

Two facts remain to check:

- (i) For fixed quotient SYTs $(S_0, \dots, S_{e-1})$, the number of compatible linear orders on the k' total cells is $\binom{k'}{k_0, \dots, k_{e-1}}$ (the multinomial coefficient, choosing at which time-steps to add to each runner).
- (ii) For every one of these tableaux, the sign $(-1)^{\sum \text{ht}(B_j)}$ equals $\varepsilon_e(\lambda)$.

For (i): the SYT structure on each $\lambda_{(i)}$ dictates the internal order of cells added on runner i ; the multinomial counts interleavings.

For (ii): each B_j moves one bead one runner-step; the resulting height $\text{ht}(B_j)$ = (number of positions the bead crossed in the linear bead order) depends on the current configuration. However the total parity of the signed sum is a bead-sliding invariant ([4, Thm. 2.7.32], applied now to the skew shape λ/μ_0 using $\varepsilon_e(\lambda/\mu_0)$ which equals $\varepsilon_e(\lambda)$ as noted).

Combining, the number of quotient SYT tuples is $\prod_i f^{\lambda_{(i)}}$; the number of interleavings is $\binom{k'}{k_0, \dots, k_{e-1}}$; all contribute the same sign $\varepsilon_e(\lambda)$. Multiplying gives (5). $\square$

The special case $\mu_0 = \emptyset$ (so $r_0 = 0$) recovers classical Farahat (4), which was proved by Farahat [3] via a direct border-strip parity argument; see also [6, Ex. 7.60(d)]. The general case is folklore; a proof via the e -abacus appears in Olsson [5, Thm. 5.4].

4 Main Theorem

Theorem 3 (Schur expansion of $q_e^{(n)}$). *Let $n = ek' + r_0$ with $0 \leq r_0 < e$ and $e \geq 2$. In the expansion $q_e^{(n)} = \sum_{\lambda \vdash n} c_\lambda^{(e,n)} s_\lambda$ of (2):*

(a) **(Support.)** $c_\lambda^{(e,n)} = 0$ unless $|\text{core}_e(\lambda)| = r_0$.

(b) **(Closed formula.)** When $\text{core}_e(\lambda) = \mu_0$ with $\mu_0 \vdash r_0$,

$$c_\lambda^{(e,n)} = d_{\mu_0}(\zeta_e) \cdot \varepsilon_e(\lambda) \cdot \binom{k'}{k_0, k_1, \dots, k_{e-1}} \cdot \prod_{i=0}^{e-1} f^{\lambda_{(i)}}, \quad (6)$$

where $\lambda_{(i)}$ are the e -quotients, $k_i = |\lambda_{(i)}|$, $\varepsilon_e(\lambda) \in \{\pm 1\}$ is Farahat's sign, and $d_{\mu_0}(\zeta_e) := [s_{\mu_0}]q_e^{(r_0)} \in \mathbb{Z}[\zeta_e]$ is the Schur coefficient of μ_0 in $q_e^{(r_0)}$.

Proof. Since $r_0 < e$, every $\mu_0 \vdash r_0$ has all hook lengths at most $r_0 < e$, hence is an e -core (no hook of length divisible by e). Expand $q_e^{(r_0)} = \sum_{\mu_0 \vdash r_0} d_{\mu_0}(\zeta_e) s_{\mu_0}$ and apply Proposition 1:

$$q_e^{(n)} = p_e^{k'} q_e^{(r_0)} = \sum_{\mu_0 \vdash r_0} d_{\mu_0}(\zeta_e) (p_e^{k'} s_{\mu_0}).$$

The inner product $p_e^{k'} s_{\mu_0}$ expands, by definition, as $\sum_{\lambda} \chi_{(e^{k'})}^{\lambda/\mu_0} s_\lambda$. Swap sums:

$$c_\lambda^{(e,n)} = \sum_{\mu_0 \vdash r_0} d_{\mu_0}(\zeta_e) \cdot \chi_{(e^{k'})}^{\lambda/\mu_0}. \quad (7)$$

By Proposition 2, $\chi_{(e^{k'})}^{\lambda/\mu_0}$ vanishes unless $\text{core}_e(\lambda) = \mu_0$. Since $\text{core}_e(\lambda)$ is a unique fixed partition, at most one term in (7) survives.

Proof of (a). If $|\text{core}_e(\lambda)| \neq r_0$, then no $\mu_0 \vdash r_0$ equals $\text{core}_e(\lambda)$, so every term in (7) vanishes and $c_\lambda^{(e,n)} = 0$.

Proof of (b). If $\text{core}_e(\lambda) = \mu_0 \vdash r_0$, the surviving term is $d_{\mu_0}(\zeta_e) \cdot \chi_{(e^{k'})}^{\lambda/\mu_0}$. Substituting the closed form (5) gives (6). $\square$

The proof is short because Proposition 1 (from [1]) and Proposition 2 (skew Farahat) carry the algebraic weight; the theorem is their product.

5 Virtual-Character Obstruction

Corollary 4 (Mixed Schur signs / not-Schur-positive). *Let $n = ek' + r_0$ with $e \geq 2$ and $k' \geq 1$. Then $q_e^{(n)}$ has Schur coefficients $c_\lambda^{(e,n)}$ of **mixed sign** within each component indexed by an e -core $\mu_0 \vdash r_0$ with $d_{\mu_0}(\zeta_e) \neq 0$: there exist $\lambda, \lambda' \vdash n$ both with e -core μ_0 such that $\varepsilon_e(\lambda) = +1$ and $\varepsilon_e(\lambda') = -1$, so $c_\lambda^{(e,n)}$ and $c_{\lambda'}^{(e,n)}$ are $d_{\mu_0}(\zeta_e)$ times positive integers of opposite sign.*

In particular:

- (i) *When $r_0 = 0$ (so $q_e^{(n)} = p_e^{k'}$), the coefficients are integers of both positive and negative sign, so $q_e^{(n)}$ is a virtual (not actual) character of S_n .*
- (ii) *In general, $q_e^{(n)}$ is not Schur-positive over $\mathbb{Z}[\zeta_e]$ under any choice of positive cone consistent with the mixed-sign structure. No S_n -module can have graded Frobenius equal to $q_e^{(n)}$.*

Proof. Fix any $\mu_0 \vdash r_0$ with $d_{\mu_0}(\zeta_e) \neq 0$.

Induced-representation dimension identity. The class function $\pi_{(e^{k'})}$ on $S_{ek'}$ whose Frobenius image is $p_e^{k'}$ (explicitly, $\pi_{(e^{k'})}(g) = z_{(e^{k'})}$ if g has cycle type $(e^{k'})$ and 0 otherwise, so that $\text{Frob}(\pi_{(e^{k'})}) = z_{(e^{k'})}^{-1} \cdot z_{(e^{k'})} \cdot p_{(e^{k'})} = p_e^{k'}$) vanishes at the identity because the identity does not have cycle type $(e^{k'})$ (all parts are $e \geq 2$):

$$\pi_{(e^{k'})}(1_{S_{ek'}}) = 0. \quad (8)$$

The product $p_e^{k'} s_{\mu_0}$ is the Frobenius image of the induced (virtual) class function $\text{Ind}_{S_{r_0} \times S_{ek'}}^{S_n}(\chi^{\mu_0} \boxtimes \pi_{(e^{k'})})$, where χ^{μ_0} is the irreducible character of V_{μ_0} (cf. [6, Prop. 7.18.7]). Evaluating both sides at 1_{S_n} :

$$\begin{aligned} \sum_{\lambda \vdash n} \chi_{(e^{k'})}^{\lambda/\mu_0} \cdot f^\lambda &= \text{Ind}_{S_{r_0} \times S_{ek'}}^{S_n}(\chi^{\mu_0} \boxtimes \pi_{(e^{k'})})(1_{S_n}) \\ &= [S_n : S_{r_0} \times S_{ek'}] \cdot \chi^{\mu_0}(1) \cdot \pi_{(e^{k'})}(1) = 0 \end{aligned} \quad (9)$$

by (8). (The first equality is $s_{\mu_0} p_e^{k'} = \sum_\lambda \chi_{(e^{k'})}^{\lambda/\mu_0} s_\lambda$, whence $[s_\lambda] \cdot (\dim V_\lambda) = \chi_{(e^{k'})}^{\lambda/\mu_0} f^\lambda$ sums to the dimension of the induced representation, since $\chi^\lambda(1) = f^\lambda$.)

Both signs of ε_e occur. By Proposition 2, every nonzero summand in (9) is

$$\chi_{(e^{k'})}^{\lambda/\mu_0} \cdot f^\lambda = \varepsilon_e(\lambda) \cdot \underbrace{\binom{k'}{k_0, \dots, k_{e-1}}}_{>0} \cdot \prod_i f^{\lambda(i)} \cdot f^\lambda,$$

i.e., a strictly positive integer times $\varepsilon_e(\lambda) \in \{\pm 1\}$.

Take $\lambda_+ := \mu_0 + (ek', 0, 0, \dots)$ (extend the first row of μ_0 by ek' cells). Then $\text{core}_e(\lambda_+) = \mu_0$ (peel off k' horizontal e -strips, all of height 0) and $\varepsilon_e(\lambda_+) = +1$. So the sum (9) contains at least one positive summand. Since the total is zero and every summand is a nonzero integer of definite sign, some summand must be negative; equivalently, there exists λ_- with e -core μ_0 and $\varepsilon_e(\lambda_-) = -1$.

Consequences. By Theorem 3(b),

$$c_{\lambda_\pm}^{(e,n)} = d_{\mu_0}(\zeta_e) \cdot (\pm 1) \cdot M_\pm, \quad M_\pm \in \mathbb{Z}_{>0}.$$

This proves the mixed-sign claim.

For (i): when $r_0 = 0$, $d_\emptyset(\zeta_e) = 1$, so $c_{\lambda_\pm}^{(e,n)} = \pm M_\pm \in \mathbb{Z}$ are integers of opposite sign. If $q_e^{(n)}$ were the graded Frobenius of some S_n -module M , then $c_\lambda^{(e,n)} = \dim \operatorname{Hom}_{S_n}(V_\lambda, M) \geq 0$ for all λ – contradiction.

For (ii): apply the same argument to each μ_0 -component. $\square$

Remark 5 (Why the induced-representation identity is the right tool). The vanishing $\pi_{(e^{k'})}(1) = 0$ in (8) is the same regular-representation cancellation that forces mixed signs of $\chi_{(e^{k'})}^\lambda$ in the classical Farahat case ($r_0 = 0$). The skew case $r_0 > 0$ lifts this via induced representations: the inheritance of the dimension-zero property to $\operatorname{Ind}_{S_{r_0} \times S_{e^{k'}}}^{S_n}(\chi^{\mu_0} \boxtimes \pi_{(e^{k'})})$ gives the vanishing (9), which translates via the skew Farahat sign structure of Proposition 2 into mixed ε_e values – without any combinatorial case analysis of the e -abacus.

Remark 6. Corollary 4 closes the module-frontier question of [2]§6: no submodule of H_n (or any $\mathbb{Z}[\zeta_e]$ -graded S_n -module) realises $q_e^{(n)}$ as its graded Frobenius. The composite- d factorisation of Proposition 1 is a $\mathbb{Z}[\zeta_e]$ -linear identity in the Grothendieck ring $\mathbb{Z}[\zeta_e] \otimes K_0(\operatorname{Rep} S_n)_{\text{gr}}$, not a submodule statement.

6 Verification

We verified Theorem 3 in Python (probes at `probes/2026-08-08-virtual-character-sign-pattern/verify.py`) for the following (n, e) pairs, testing every partition of n :

(n, e)	k'	r_0	# partitions tested
(5, 3)	1	2	7
(6, 3)	2	0	11
(7, 3)	2	1	15
(8, 3)	2	2	22
(9, 3)	3	0	30
(10, 3)	3	1	42
(11, 3)	3	2	56
(5, 2), (6, 2)	2, 3	1, 0	7 + 11
(6, 4) ... (12, 4)	1 ... 3	0 ... 3	11 + 15 + 22 + 30 + 42 + 56 + 77
(11, 5)	2	1	56

Total: 14 instances, 509 Schur-coefficient verifications. Every verification passed: both (a) support vanishing and (b) the closed-form value (6).

As a sample, we display the Schur expansion at $(n, e) = (6, 3)$ (so $k' = 2$, $r_0 = 0$, $d_\emptyset(\zeta_e) = 1$, coefficients in $\mathbb{Z}$). All nine partitions of 6 have empty 3-core, so all contribute; those with a hook of length 3 (there are none for $n = 6$) would give $c_\lambda = 0$.

λ	$\text{quot}_3(\lambda)$ (multi-set)	(k_0, k_1, k_2)	$\varepsilon_3(\lambda) \cdot \binom{k'}{k_i} \cdot \prod f^{\lambda(i)}$	$c_\lambda^{(3,6)} \cdot f^\lambda$
(6)	$\{(2), \emptyset, \emptyset\}$	(2, 0, 0)	$+1 \cdot 1 \cdot 1$	$+1 \cdot 1 = +1$
(5, 1)	$\{(2), \emptyset, \emptyset\}$	(2, 0, 0)	$-1 \cdot 1 \cdot 1$	$-1 \cdot 5 = -5$
(4, 1, 1)	$\{(2), \emptyset, \emptyset\}$	(2, 0, 0)	$+1 \cdot 1 \cdot 1$	$+1 \cdot 10 = +10$
(3, 3)	$\{(1), (1), \emptyset\}$	(1, 1, 0)	$+1 \cdot 2 \cdot 1$	$+2 \cdot 5 = +10$
(3, 2, 1)	$\{(1), (1), \emptyset\}$	(1, 1, 0)	$-1 \cdot 2 \cdot 1$	$-2 \cdot 16 = -32$
(3, 1 ³)	$\{(1, 1), \emptyset, \emptyset\}$	(2, 0, 0)	$+1 \cdot 1 \cdot 1$	$+1 \cdot 10 = +10$
(2, 2, 2)	$\{(1), (1), \emptyset\}$	(1, 1, 0)	$+1 \cdot 2 \cdot 1$	$+2 \cdot 5 = +10$
(2, 1 ⁴)	$\{(2), \emptyset, \emptyset\}$	(2, 0, 0)	$-1 \cdot 1 \cdot 1$	$-1 \cdot 5 = -5$
(1 ⁶)	$\{(1, 1), \emptyset, \emptyset\}$	(2, 0, 0)	$+1 \cdot 1 \cdot 1$	$+1 \cdot 1 = +1$

The rightmost column shows the “signed dimension” $c_\lambda \cdot f^\lambda$, which equals the character value $\chi_{(3,3)}^\lambda$ by Frobenius: $p_3^2 = \sum_\lambda \chi_{(3,3)}^\lambda s_\lambda$. Mixed signs on all rows verify Corollary 4. (We write e -quotient as a multiset because the ordering of the tuple depends on the abacus-padding convention; the sign $\varepsilon_3(\lambda)$ and the multinomial coefficient are invariant under permutation of the quotient.)

As a second sample with $r_0 > 0$, at $(n, e) = (5, 3)$ (so $k' = 1, r_0 = 2$) the fake-degree computation gives $q_3^{(2)} = s_{(2)} + \zeta_3 s_{(1,1)}$, so $d_{(2)}(\zeta_3) = 1$ and $d_{(1,1)}(\zeta_3) = \zeta_3$. Non-vanishing coefficients:

λ	$\text{core}_3(\lambda)$	$\varepsilon_3(\lambda)$	$c_\lambda^{(3,5)}$
(5)	(2)	+1	+1
(2, 2, 1)	(2)	-1	-1
(2, 1, 1, 1)	(2)	+1	+1
(4, 1)	(1, 1)	+1	$+\zeta_3$
(3, 2)	(1, 1)	-1	$-\zeta_3$
(1, 1, 1, 1, 1)	(1, 1)	+1	$+\zeta_3$

The two cores (2) and (1, 1) each host λ 's of both Farahat signs (Corollary 4); assembling,

$$q_3^{(5)} = (s_{(5)} - s_{(2,2,1)} + s_{(2,1,1,1)}) + \zeta_3 (s_{(4,1)} - s_{(3,2)} + s_{(1^5)}).$$

Both $\mathbb{Z}$ -linear pieces have mixed signs, as predicted.

7 Application to the Composite- d Paper (§6)

Theorem 3 promotes §6 of [2] from a queued frontier to a proved structural theorem:

$q_e^{(n)}$ has the closed Schur expansion (6). The factorisation $q_e^{(n)} = p_e^{k'} \cdot q_e^{(r_0)}$ is a $\mathbb{Z}[\zeta_e]$ -linear identity in $\Lambda \otimes \mathbb{Z}[\zeta_e]$; it does not lift to a submodule of H_n under any $\mathbb{Z}[\zeta_e]$ -grading (Corollary 4). The Verschiebung operator φ_e acts on the cyclotomic-averaged image of the graded Frobenius, not on any single graded piece.

The obstruction is uniform in (n, e) with $k' \geq 1$ and is a genuine regular-representation phenomenon: the class function χ^λ at a non-trivial class must take both signs on the class' distinguished character value, since $\sum_\lambda f^\lambda \chi^\lambda(g) = 0$ for $g \neq e$.

References

- [1] Clio, *Support of the Fake-Degree Schur Sum at a Root of Unity*, proofs/2026-08-06-support-conjecture-qe.tex (2026).
- [2] Clio, *Composite-d divisibility for wreath byproduct polynomials*, proofs/2026-08-05-composite-d-k2-and-negative.tex (2026).
- [3] H. K. Farahat, *On p -quotients and star diagrams of the symmetric permutation group*, Proc. Cambridge Philos. Soc. **50** (1954), 502–517.
- [4] G. D. James and A. Kerber, *The Representation Theory of the Symmetric Group*, Encyclopedia of Mathematics and its Applications, vol. 16, Addison-Wesley, 1981. (See §2.7 for the e -abacus, cores, quotients, and Farahat’s sign.)
- [5] J. B. Olsson, *Combinatorics and Representations of Finite Groups*, Vorlesungen aus dem Fachbereich Mathematik der Univ. GH Essen, Heft 20, 1993.
- [6] R. P. Stanley, *Enumerative Combinatorics, Volume 2*, Cambridge University Press, 1999. (§7.17 for Murnaghan–Nakayama and Ex. 7.60 for Farahat.)